16	A beam of plane-polarised light of intensity I falls normally onto a thin sheet of polaroid. If the transmitted beam has an intensity of $\frac{3}{4}I$, what is the angle between the plane of polarisation of the incident beam and the polarising axis of the polaroid?						
	A	30°	**B**	41°	**C**	45°	**D** 60°
	L2	**Answer: A** Using Malus' Law, $I_{transmitted} = I_0\cos^2\theta$ $\frac{3}{4}I = I\cos^2\theta$ $\cos^2\theta = \frac{3}{4}$ $\cos\theta = \frac{\sqrt{3}}{2}$ $\theta = 30°$					

17 A hemispherical bowl of radius 200 mm is completely filled with water at rest. If its side is